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Feature Tested: Authentication Module (Registration, Login, Password Reset)

1. Overview

This document provides evidence that all unit tests for the **Authentication Module** were executed successfully and that the results were documented using a proper automated reporting tool(mochawesome), as required by the assignment rubric.

The following API endpoints were tested:

- **POST /api/users/register**
- **POST /api/users/login**
- **POST /api/users/request-password-reset**
- **POST /api/users/reset-password**

The 3 test files include the following :

Registration Tests Summary (**registration.test.js**)

- Valid registration creates a new user, stores hashed password, and triggers a welcome email (stubbed).
- Duplicate email/username returns 400 with appropriate error.
- Weak password is rejected and no user is saved.

Login Tests Summary (**login.test.js**)

- Valid login returns a success response and triggers loginAlert (stubbed).
- Incorrect password returns 400 error.
- Logging in with a non-existent user returns 404.

Password Reset Tests Summary (**passwordReset.test.js**)

- Reset code generation stores a reset token and sends an email (stubbed).
- Valid reset code allows password update.
- Invalid reset code returns an error without changing data.

All tests were executed using:

- **Mocha** – test runner
- **Chai** – assertion library
- **Supertest** – HTTP/API request testing
- **Sinon** – mocking & stubbing emails
- **Mochaawesome** – HTML/JSON reporting

2. How the Tests Were Executed

Just for testing(no report):

Use :

npm run test:auth

For generating a mochaawesome report for all at once:

Use:

npm run report:auth

Note:

Mochawesome report will be generated on the

report:auth script (package.json)

```
"report:auth": "cross-env NODE_ENV=test mocha TESTS/registration.test.js  
TESTS/login.test.js TESTS/passwordReset.test.js --reporter mochawesome  
--reporter-options reportDir=TESTS/nirajanbanjade-report,reportFilename=auth-report  
--timeout 10000"
```

This part in package.json has described the reporting directory as nirajanbanjade-report inside TESTS/.

So, npm run report:auth will generate a report inside TESTS.

This script performs the following actions:

1. Sets NODE_ENV=test so emails and side-effects are disabled.
2. Runs all three authentication test files at once.
3. Uses Mochawesome reporter to generate HTML/JSON reports.
4. Saves the results in a reporting folder.

